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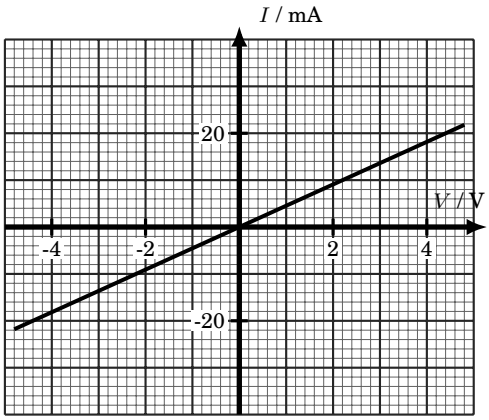
Date: _____

Mr. Rawson • GCSE Physics

Ohmic and Non-Ohmic Conductors

Three components, three shapes. Read the shape and you know the component.

1 · Ohmic conductors resistance stays the same



Graph A — a fixed resistor

The shape: a **straight line through the origin**.

Double the pd, and the current doubles too. So $R = \frac{V}{I}$ comes out **the same everywhere on the line**.

$$I \propto V$$

Examples

- a **fixed resistor**
- a **metal wire** — as long as it stays cool

The condition: *at a constant temperature.* Let it get hot and it stops being ohmic.

Worked example — the method

A point read off a *different* resistor's graph gives $V = 3.0\,\text{V}$ and $I = 25\,\text{mA}$.

Step 1 — convert the current to amps. $25\,\text{mA} = 25 \div 1000 = 0.025\,\text{A}$

Step 2 — divide. $R = \frac{V}{I} = \frac{3.0}{0.025} = \mathbf{120}\,\Omega$

Forgetting step 1 gives 0.12 — a thousand times too small. The current axis on this sheet is always in **milliamps**.

Finding resistance from any graph

Pick a point on the line. Read V across. Read I up. Divide. $R = \frac{V}{I}$ — this works on *every* graph on this sheet.

The check: do it again at a second point. Same answer → the resistance is **constant** (ohmic). Different answer → the resistance has **changed**.

Exam-style question [4 marks]

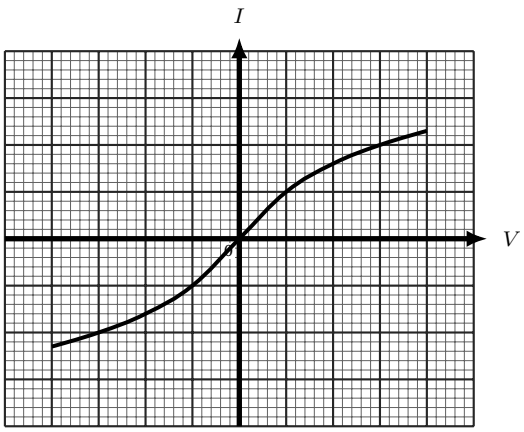
Graph A is a real fixed resistor. Use the graph to find its resistance.

Choose a point where the line crosses a grid line *exactly* — it makes the reading easier.

The point I read: $V =$ _____ V $\quad I =$ _____ mA

2 · Non-ohmic conductors resistance changes as you use it

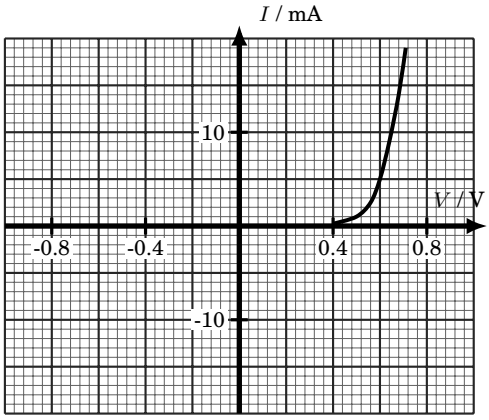
The resistance of a **lamp**, **diode**, **thermistor** or **LDR** is **not** constant — so their graphs are not straight lines.



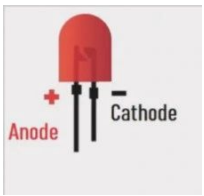
Graph B



the filament and its support wires



Graph C — a diode
note the scale: 1 square = 0.2 V



an LED is a diode — it only works one way round

2a The filament lamp

The shape: bends over towards the V axis — the same way on *both* sides.

Why: bigger current → filament **hotter** → ions **vibrate more** → more **collisions** → **resistance increases**.

Examples: a torch bulb, a car headlamp, an old-style light bulb.

Exam-style question [3 marks]

Explain why the resistance of a filament lamp increases as the potential difference across it increases.

2b The diode

The shape: flat along the axis one way, then rising steeply the other way.

Why: a diode lets current through in **one direction only**. Forwards, it conducts once the pd passes about 0.6 V. Backwards, its resistance is **very high**, so the current is almost zero.

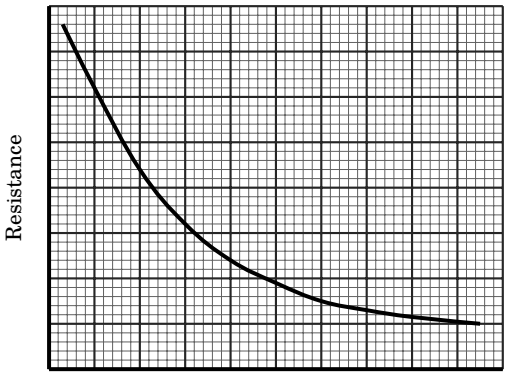
Examples: an **LED** is a diode — connect it the wrong way round and it simply will not light. Also the charging circuit in a phone, a solar panel, and anything that must not be connected backwards.

Exam-style question [2 marks]

A diode is connected in a circuit with a cell and a lamp, and the lamp lights. The diode is then turned round. Describe and explain what happens to the lamp.

Name: _____ **Date:** _____

Both components below are non-ohmic too — but their graphs use **different axes**. Instead of I against V, we plot **resistance** against the thing the component responds to.

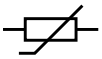


Graph D



the black bead at the top is the temperature sensor

2c The thermistor



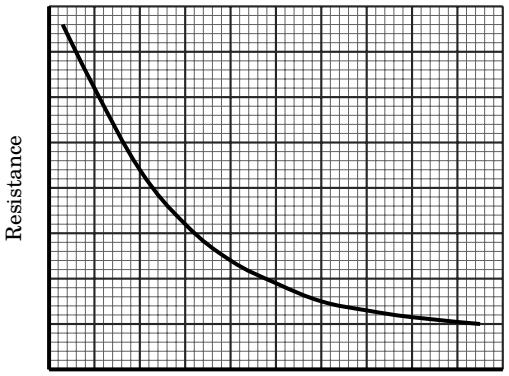
The shape: resistance **falls** as the temperature **rises** — steeply at first, then levelling off.

Remember it as: *hotter → lower resistance → bigger current.*

Examples: a thermostat, a fire alarm, the temperature sensor in an oven or a car engine.

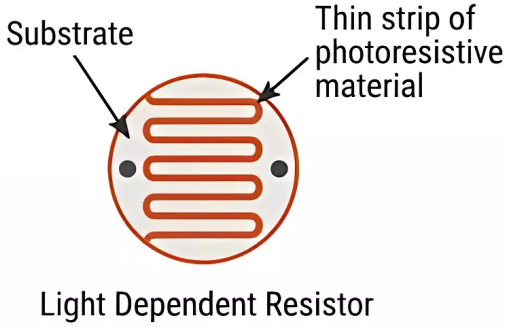
Exam-style question [3 marks]

A thermistor is connected in series with a lamp and a cell. Describe and explain what happens to the brightness of the lamp as the room gets warmer.



Graph E

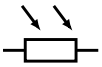




Light Dependent Resistor

the track is the part that responds to light

2d The LDR



LDR stands for **light dependent resistor**.

The shape: the *same* falling curve as the thermistor — but against **light intensity**. Resistance **falls** as it gets **brighter**.

Remember it as: *brighter → lower resistance → bigger current.*

Examples: street lights that come on at dusk, automatic camera exposure, a phone screen that dims in a dark room.

Exam-style question [3 marks]

Explain how an LDR can be used to switch a street light on automatically when it gets dark.

Name: _____ **Date:** _____

3 · Questions graphs A, B and C are on the front

1. State what is meant by an **ohmic conductor**. **[2]**

2. Name the component that produced each graph. **[3]**

   A: **B:** **C:**

3. Use **graph B** to find the resistance of the lamp at $V = 1.0\,\text{V}$ and at $V = 3.0\,\text{V}$. Show your working, and say what has happened to the resistance. **[4]**

4. Jamie found the resistance for **graph A** by measuring its gradient, and wants to use the same method on **graph B**. State what Jamie should do instead for graph B, and explain why the method that worked on graph A does not carry across. **[3]**

5. Explain, in terms of the filament, **why** graph B bends. **[3]**

6. Look at the **left-hand half** of the graphs, where the pd has been reversed. **[4]**

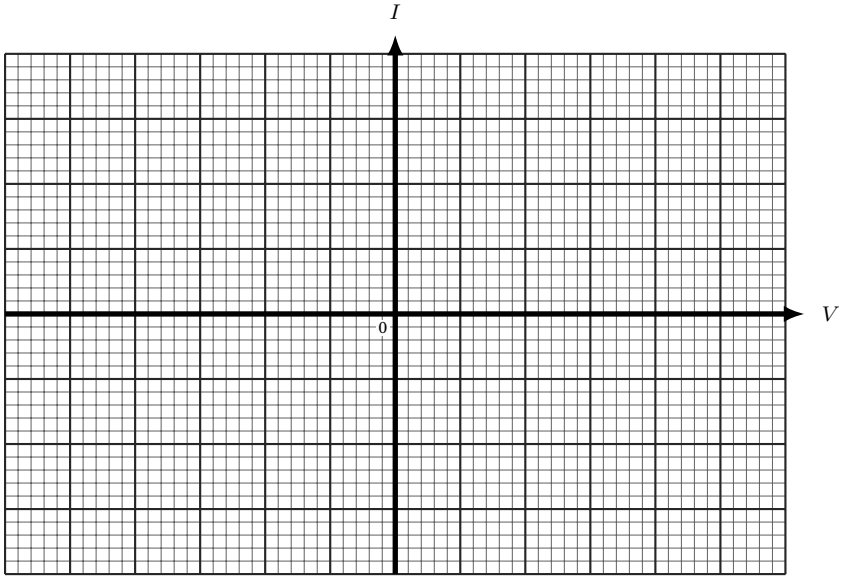
   (a) Describe what **graph C** does, and explain why.

   (b) Graph B is symmetrical about the origin. What does that tell you about a filament lamp?

7. A fixed resistor is left switched on until it becomes noticeably warm, and the test is repeated. Circle what happens to its graph, and explain in one sentence. **[2]**

   it begins to bend over it stays perfectly straight it flattens along the axis

8. Sketch the graph you would expect for a **filament lamp**. Use both halves of the axes. **[3]**



Filament lamp photograph: KMJ via Wikimedia Commons, CC BY-SA 3.0.